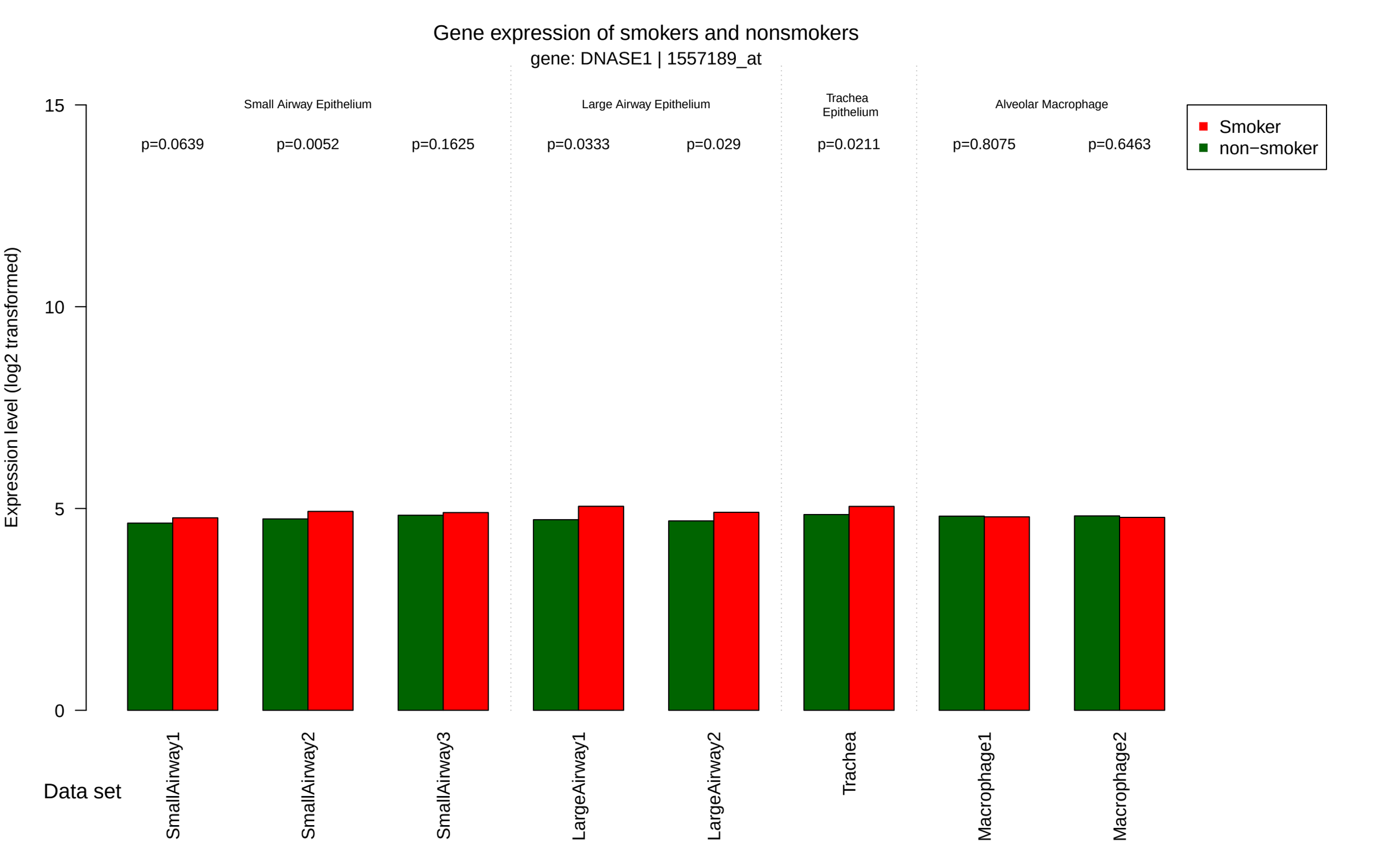
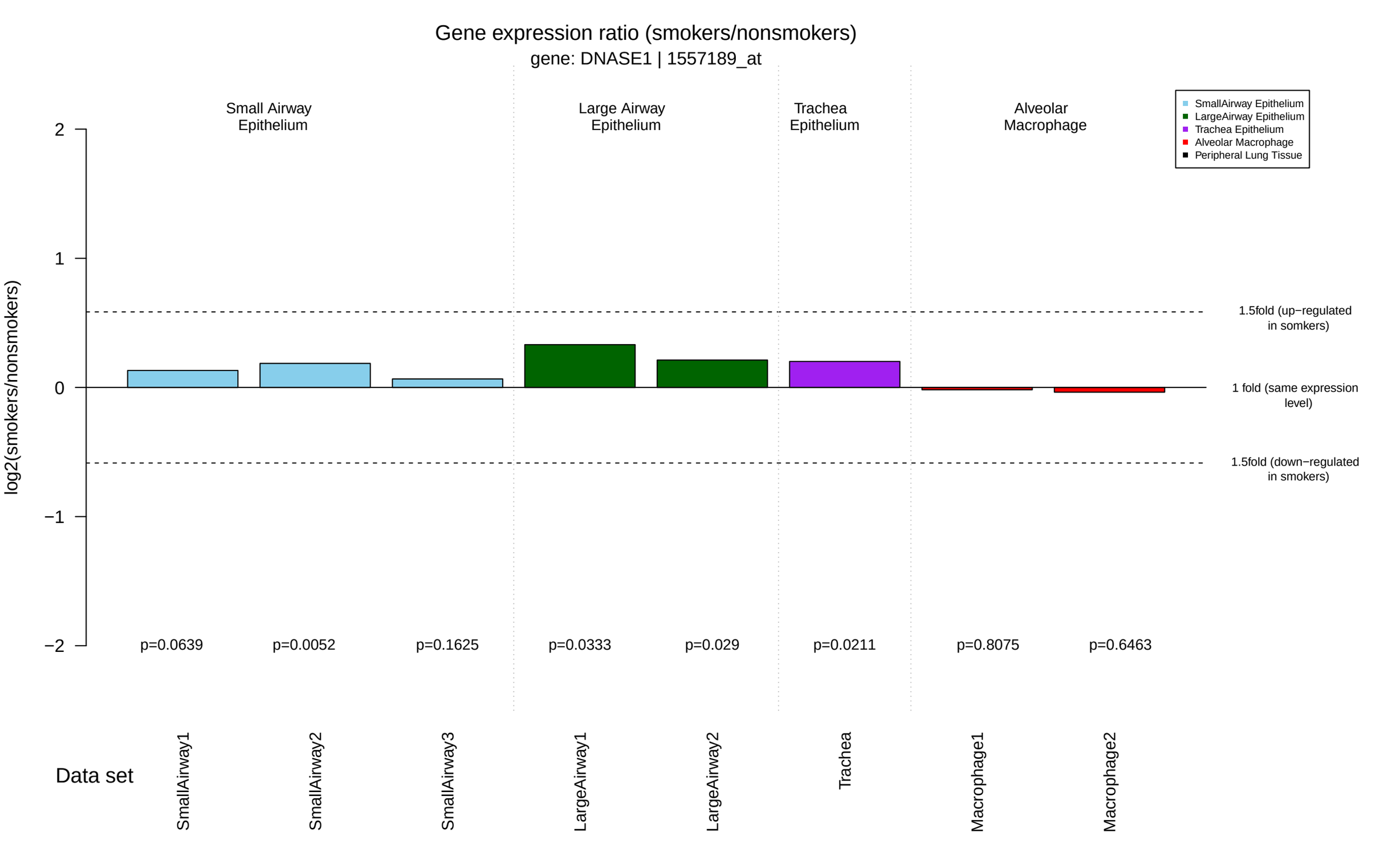
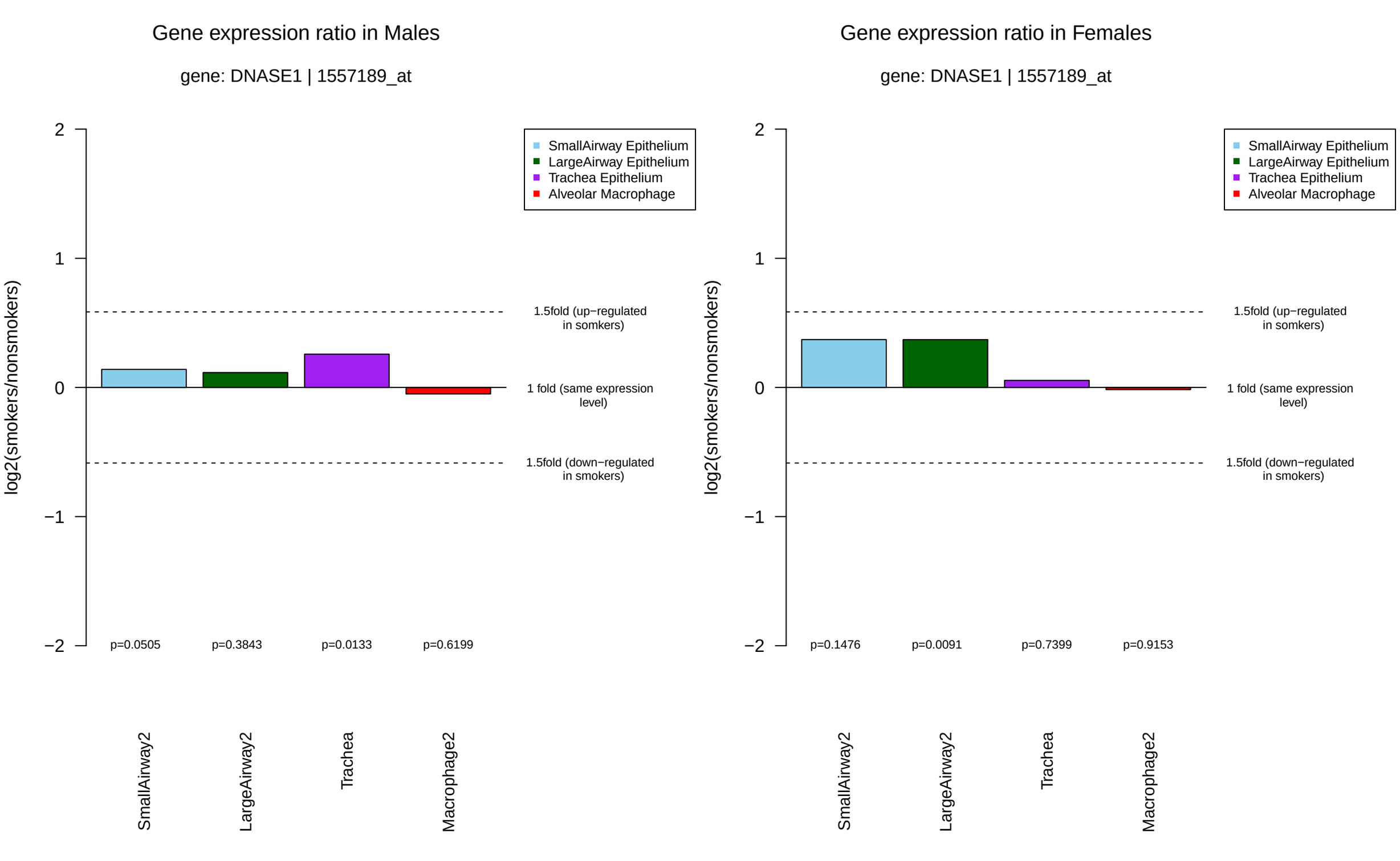
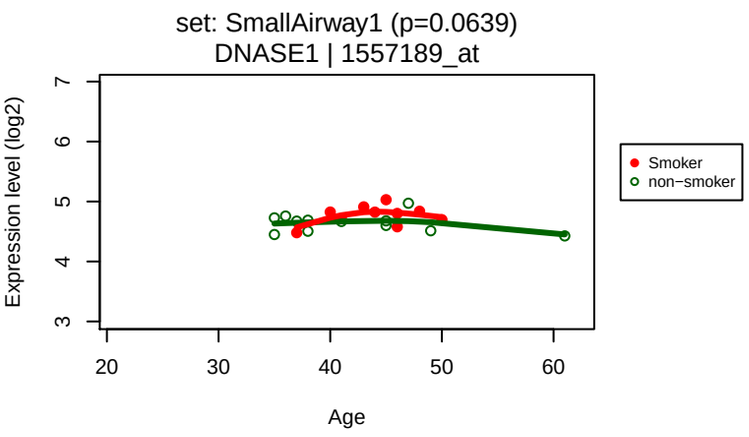
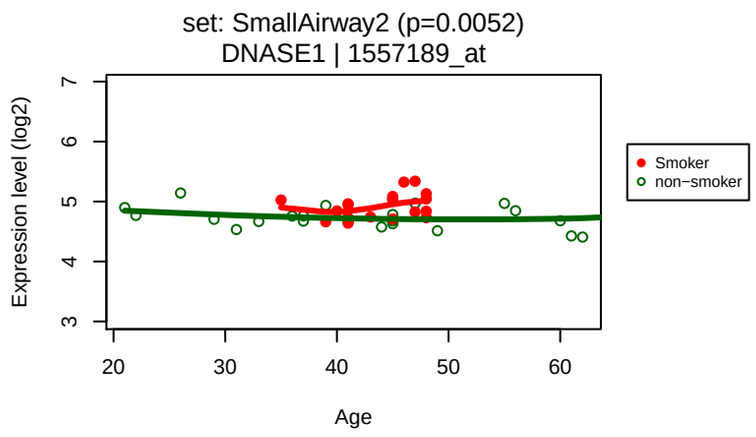
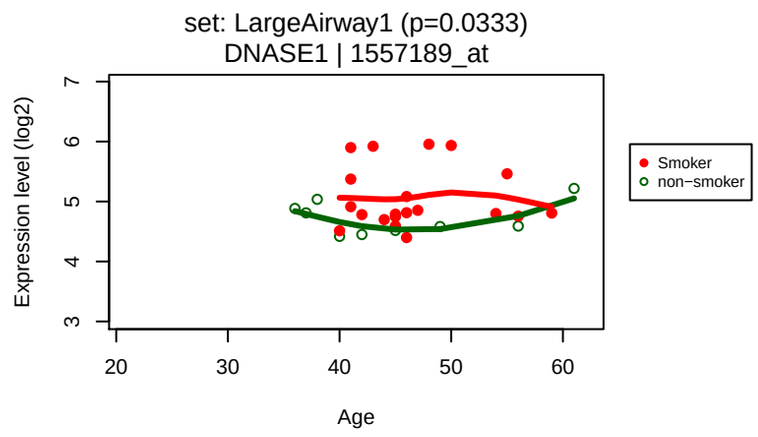

Gene expression of smokers and nonsmokers
gene: DNASE1 | 1557189_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0639
p=0.0052
p=0.1625
p=0.0333
p=0.029
p=0.0211
p=0.8075
p=0.6463
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

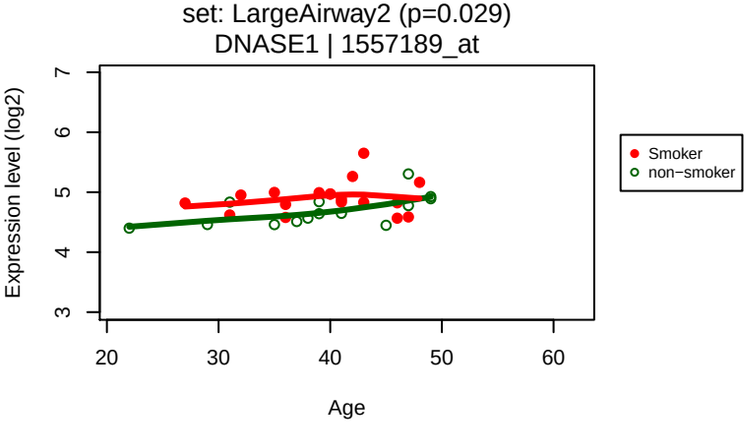

Gene expression ratio (smokers/nonsmokers)
gene: DNASE1 | 1557189_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0639
p=0.0052
p=0.1625
p=0.0333
p=0.029
p=0.0211
p=0.8075
p=0.6463
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

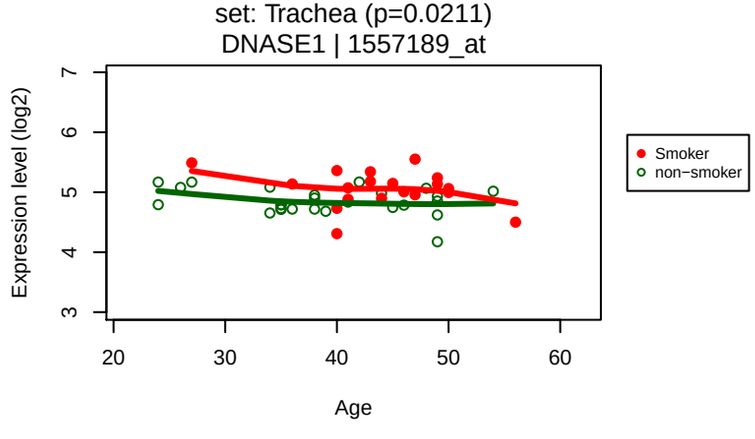

Gene expression ratio in Males
gene: DNASE1 | 1557189_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0505
p=0.3843
p=0.0133
p=0.6199
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNASE1 | 1557189_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1476
p=0.0091
p=0.7399
p=0.9153
SmallAirway2
LargeAirway2
Trachea
Macrophage2

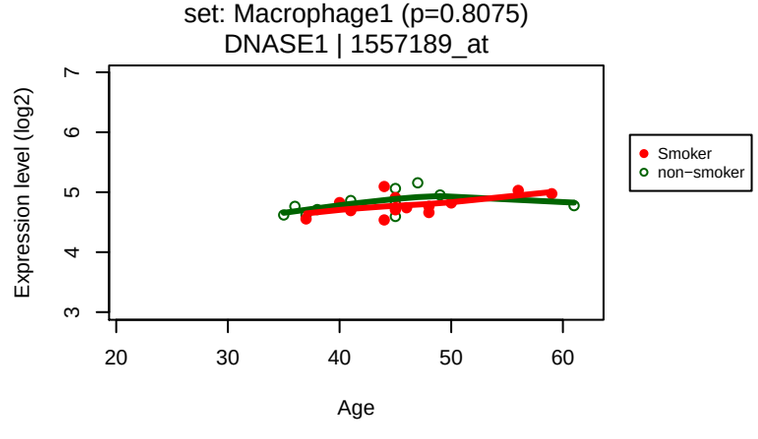

set: SmallAirway1 (p=0.0639)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker

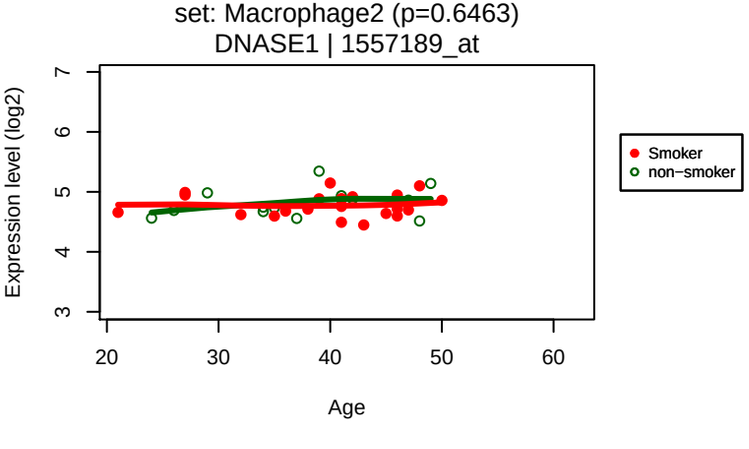

set: SmallAirway2 (p=0.0052)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker

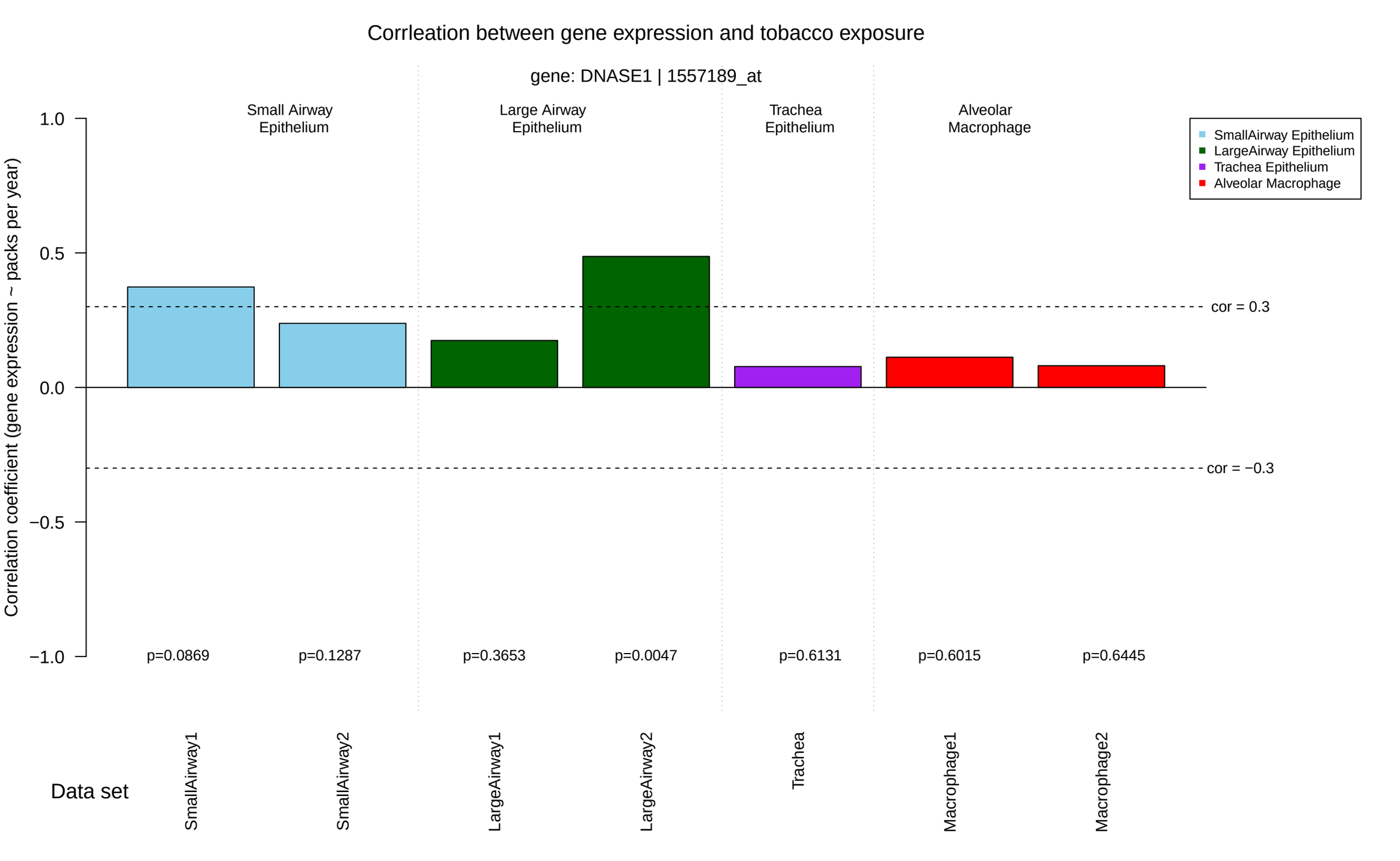

set: LargeAirway1 (p=0.0333)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.029)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0211)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8075)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6463)
DNASE1 | 1557189_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNASE1 | 1557189_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0869
p=0.1287
p=0.3653
p=0.0047
p=0.6131
p=0.6015
p=0.6445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2